# SCIENCE MASTER X

Complete Academic Study Notes • Class 10 Science

## CHAPTER 12

# Magnetic Effects of Electric Current

(Full Chapter)

> Magnetic Field & Field Lines • Oersted's Experiment • Field due to Straight Conductor, Circular Loop & Solenoid • Right-Hand Thumb Rule • Electromagnet • Force on Current-Carrying Conductor • Fleming's Left-Hand Rule • Domestic Electric Circuits

**Target Level: CBSE Class 10**

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## 📖 Chapter Overview & Basic Concepts

An electric current flowing through a conductor produces a magnetic field around it. This fundamental phenomenon links electricity and magnetism, giving rise to electromagnetism.

SECTION 1: MAGNETIC FIELD & FIELD LINES

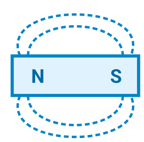
### 1.1 Magnet & Magnetic Field

- **Magnet:** An object that attracts materials like iron, steel, nickel, and cobalt. Every magnet has two poles: **North (N)** and **South (S)**.
- **Like poles repel** each other, whereas **unlike poles attract** each other.
- **Magnetic Field ($B$):** The space around a magnet in which its magnetic influence can be detected. It is a vector quantity (has both magnitude and direction). SI Unit: **Tesla (T)**.

### 1.2 Properties of Magnetic Field Lines

- Field lines emerge from the **North pole** and merge at the **South pole** outside the magnet. Inside the magnet, they move from South to North pole (forming closed continuous loops).
- The relative strength of the field is shown by the degree of closeness of the field lines (crowded lines indicate a stronger field).
- **No two field lines intersect each other:** If they did, it would mean that at the point of intersection, a compass needle would point in two different directions simultaneously, which is impossible.

📌 **Diagram: Magnetic Field Lines Around a Bar Magnet**




**NCERT High-Yield Insight:** Christian Oersted (1820) discovered electromagnetism accidentally when he noticed that an electric current flowing through a copper wire deflected a nearby compass needle.

SECTION 2: MAGNETIC FIELD DUE TO CURRENT-CARRYING CONDUCTORS

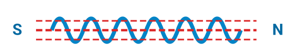
### 2.1 Straight Conductor & Right-Hand Thumb Rule

- The magnetic field lines around a straight current-carrying wire form **concentric circles** centered on the wire.
- **Right-Hand Thumb Rule:** Imagine holding a current-carrying straight conductor in your right hand such that the thumb points in the direction of current. Then, your fingers wrapped around the conductor will point in the direction of the magnetic field lines.

### 2.2 Circular Loop & Solenoid

- **Circular Loop:** At every point of a circular loop, the field lines are concentric circles. At the center of the loop, the field lines appear as straight lines. Strength increases with number of turns ($N$).
- **Solenoid:** A coil of many circular turns of insulated copper wire wrapped closely in the shape of a cylinder.
- The magnetic field pattern of a solenoid is identical to that of a **bar magnet**. One end acts as N-pole and the other as S-pole.
- The field inside a long solenoid is **uniform** (parallel straight lines).

📌 **Diagram: Magnetic Field Lines Through a Solenoid**




Uniform Magnetic Field Inside

### 2.3 Electromagnet vs Permanent Magnet

| Property | Electromagnet | Permanent Magnet |
|---|---|---|
| **Core Material** | Soft iron core placed inside a solenoid | Steel or alloy (e.g., Alnico) |
| **Magnetism Nature** | Temporary (loses magnetism when current is off) | Permanent (cannot easily be demagnetized) |
| **Strength** | Can be changed by varying current or turns | Fixed strength |
| **Polarity** | Can be reversed by reversing current direction | Fixed polarity |

## SECTION 3: FORCE ON A CURRENT-CARRYING CONDUCTOR

### 3.1 Force in a Magnetic Field

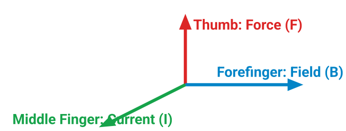
- A current-carrying conductor experiences a mechanical force when placed in an external magnetic field.
- The direction of force is reversed if the direction of current or magnetic field is reversed.
- **Maximum Force:** Experienced when the direction of current is **perpendicular ($90^\circ$)** to the magnetic field.
- **Zero Force:** When current flows parallel or antiparallel to the magnetic field ($0^\circ$ or $180^\circ$).

### 3.2 Fleming's Left-Hand Rule

Stretch the thumb, forefinger, and middle finger of your left hand mutually perpendicular to each other:

**Forefinger** → Direction of Magnetic Field ($B$)
**Middle Finger** → Direction of Electric Current ($I$)
**Thumb** → Direction of Motion / Force ($F$)

📌 **Diagram: Fleming's Left-Hand Rule**




**NCERT High-Yield Insight:** Electric motors, loudspeakers, microphones, and measuring instruments (like galvanometers) work on the principle of mechanical force experienced by a current-carrying conductor in a magnetic field.

SECTION 4: DOMESTIC ELECTRIC CIRCUITS

### 4.1 Key Specifications & Wire Color Codes

- Electric power in homes is supplied through mains at **220 V** AC with a frequency of **50 Hz**.

| Wire Type | Insulation Color Code | Function / Potential Difference |
|---|---|---|
| **Live Wire ($L$)** | Red / Brown | Carries high voltage ($+220\text{ V}$). Positive supply line. |
| **Neutral Wire ($N$)** | Black / Blue | Completes circuit ($0\text{ V}$). Potential difference $V_{L-N} = 220\text{ V}$. |
| **Earth Wire ($E$)** | Green / Yellow | Connected to a metal plate deep in Earth. Provides low resistance safety path for leakage current. |

### 4.2 Circuit Hazards & Safety Devices

- **Short Circuiting:** Occurs when live wire and neutral wire come in direct contact due to damaged insulation or fault in appliances, causing current to increase abruptly.
- **Overloading:** Occurs when too many electrical appliances of high power rating are operated simultaneously on a single socket, drawing current beyond line capacity.
- **Electric Fuse:** Safety wire with low melting point placed in *series with live wire*. Protects appliances against overloading and short circuiting by melting during excessive current flow.
- **Earthing:** Metallic body of appliances (e.g., refrigerator, toaster) is connected to earth wire. Prevents severe electric shock to user in case of current leakage.

## ❓ Important Practice Questions

**Q1: Why do two magnetic field lines never intersect each other?**
**Ans:** If they intersect, it implies that at the point of intersection, the magnetic field has two different directions, which is physically impossible.

**Q2: State Fleming's Left-Hand Rule. Where is it applied?**
**Ans:** Stretch thumb, forefinger, and middle finger of left hand mutually perpendicular. If forefinger points to magnetic field and middle finger to current, thumb gives the direction of force. It is used to determine force on current-carrying wire in electric motors.

**Q3: What is the function of an earth wire in domestic circuits? Why is earthing necessary for metallic appliances?**
**Ans:** Earth wire provides a low resistance path for current to flow safely into Earth. Earthing prevents fatal electric shocks if live wire accidentally touches the metal casing of appliances.

**Q4: What is the frequency of AC power supply in India? How many times does its direction change per second?**
**Ans:** The frequency of AC supply in India is **50 Hz**. Since AC changes direction twice per cycle, its direction changes **100 times per second**.